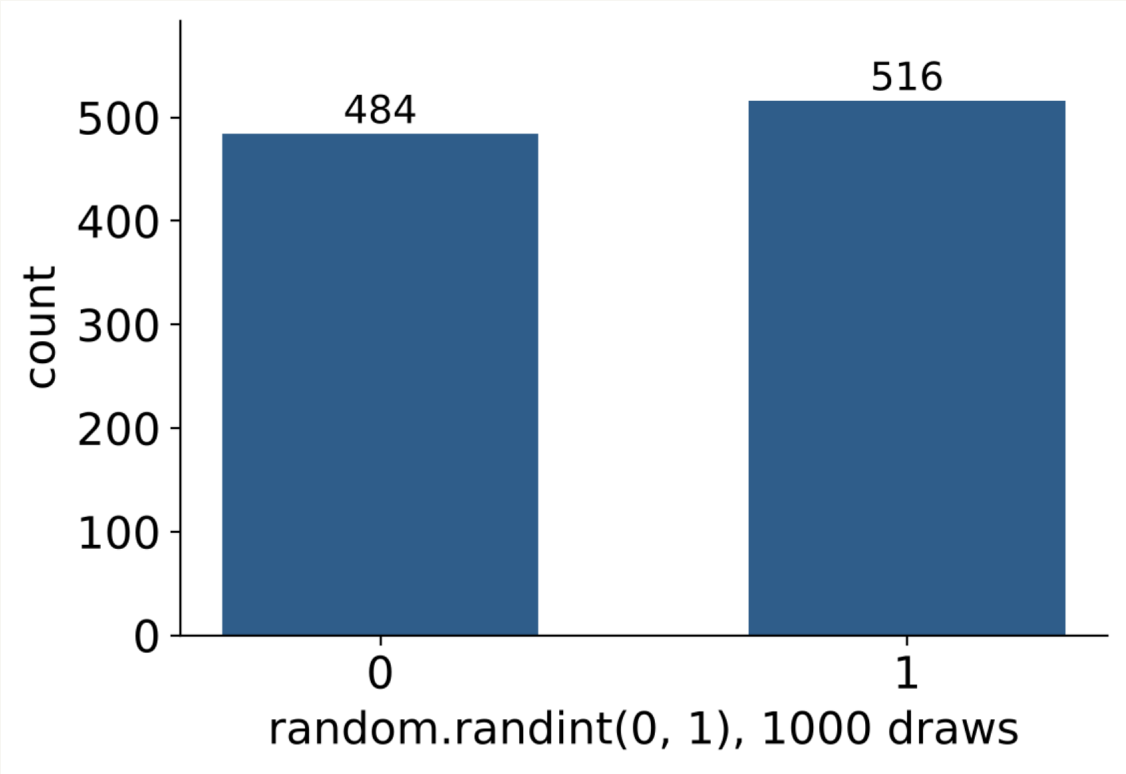
Quantum Hello World

- One project: a random number generator
- Six steps, one circuit each
- Slides first, then the notebook on Kaggle

UMKC Quantum Computing Workshop, October 24, 2026

Why a random number generator

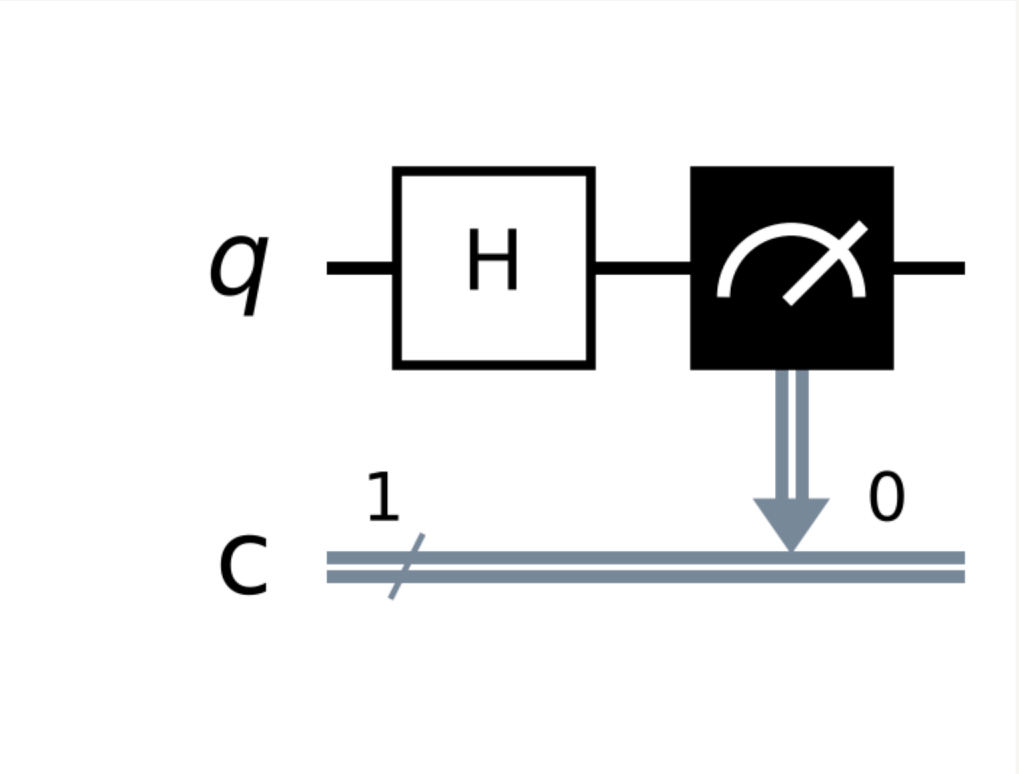
- Encryption keys, lotteries, and simulations need numbers nobody can predict
- A classical computer computes its random numbers, so they are predictable
- A qubit is not computed. It is read.



Section 1 of the notebook: Python random.randint, 1000 draws

What a circuit is made of

- Qubits: the horizontal lines. Every qubit starts at 0.
- Gates: the boxes. They change what the qubit will give when read.
- Measurements: the meter. Reading a qubit always gives a 0 or a 1.



QuantumCircuit(1, 1) then .h(0) then .measure(0, 0)

The H gate: one random bit

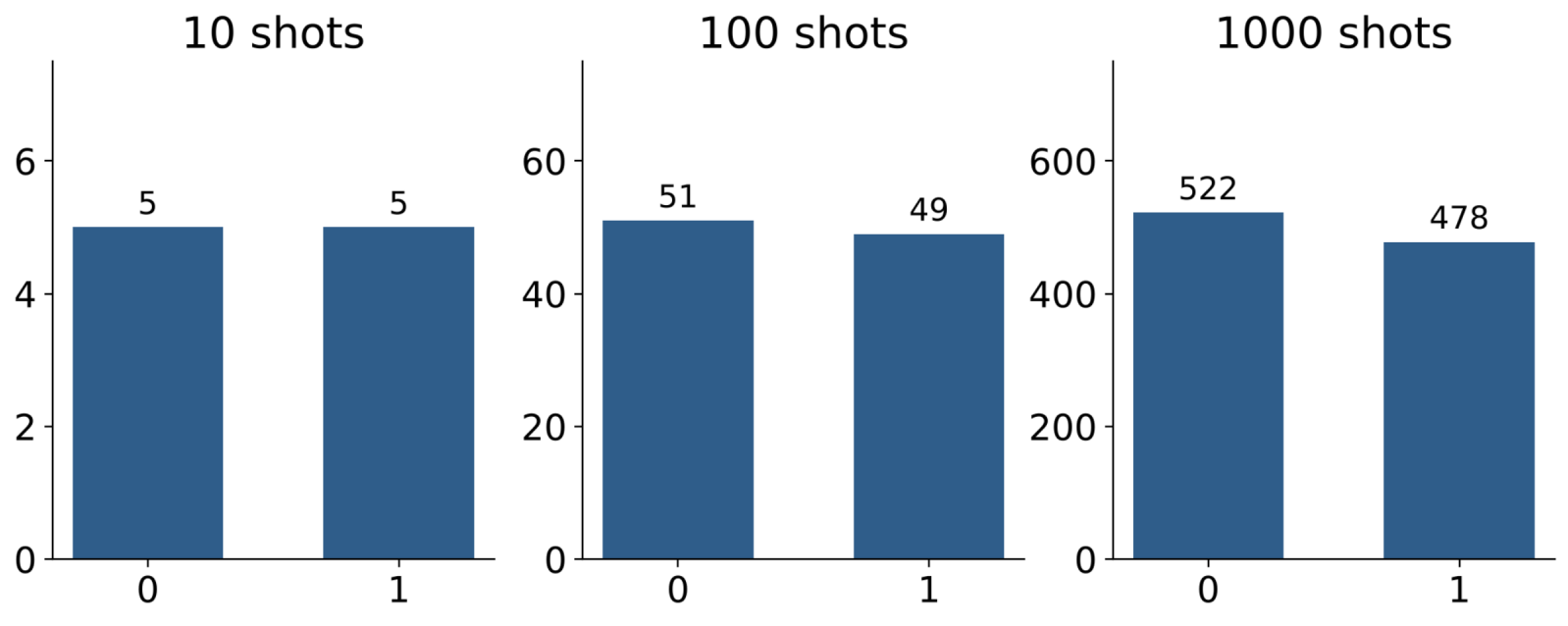
- H on a qubit at 0 makes 0 and 1 equally likely when read
- One run of the circuit gives one bit
- Run it 1000 times and count. Same histogram as Python, different source.



Section 2 of the notebook

Shots

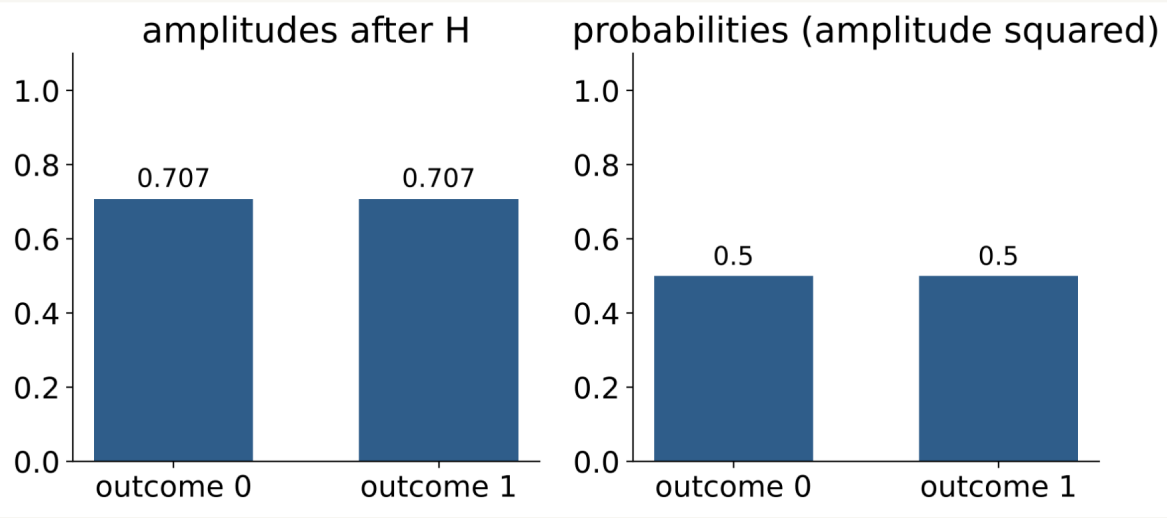
- A shot is one run of the circuit, start to measurement
- Few shots wobble. Many shots settle on the true odds.
- Every result in the notebook is a count over shots



Exercise 2 in the notebook: drop shots to 10 and see how uneven it gets

Where the odds come from: amplitudes

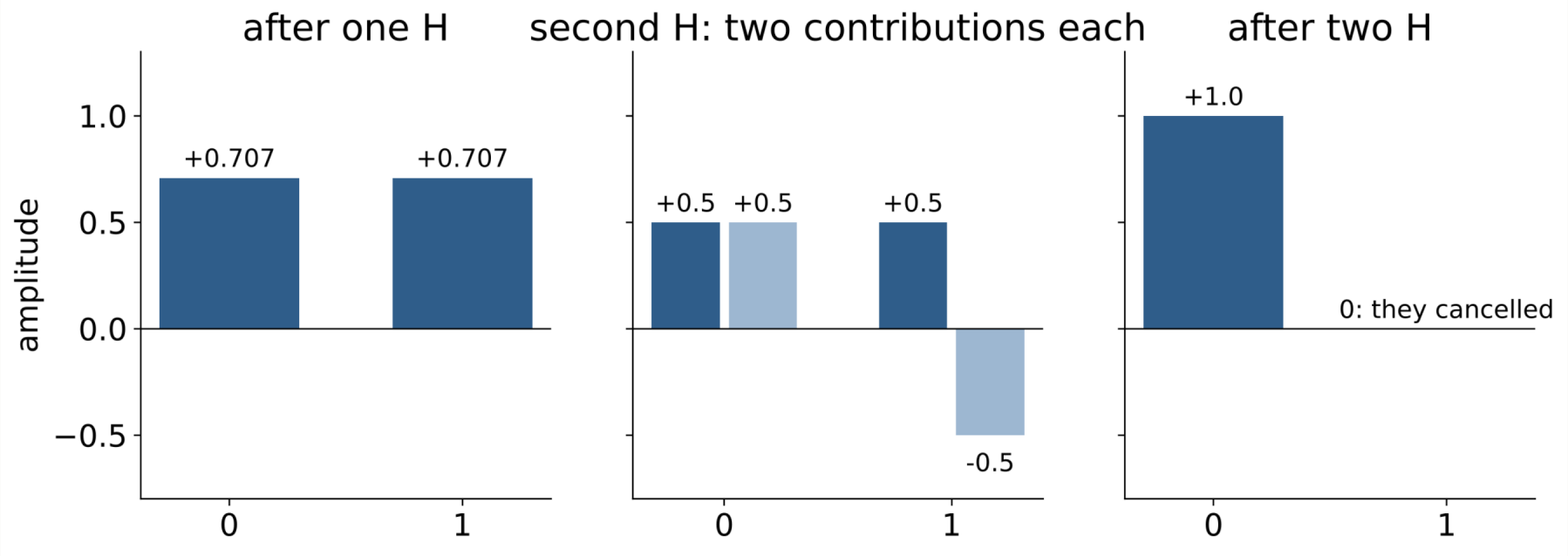
- A qubit’s state is two numbers, one for outcome 0 and one for outcome 1
- Probability of an outcome is its number squared
- After H both numbers are 0.707, and 0.707 squared is 0.5



Section 3 of the notebook: Statevector(circuit), simulator only

H twice: randomness undone

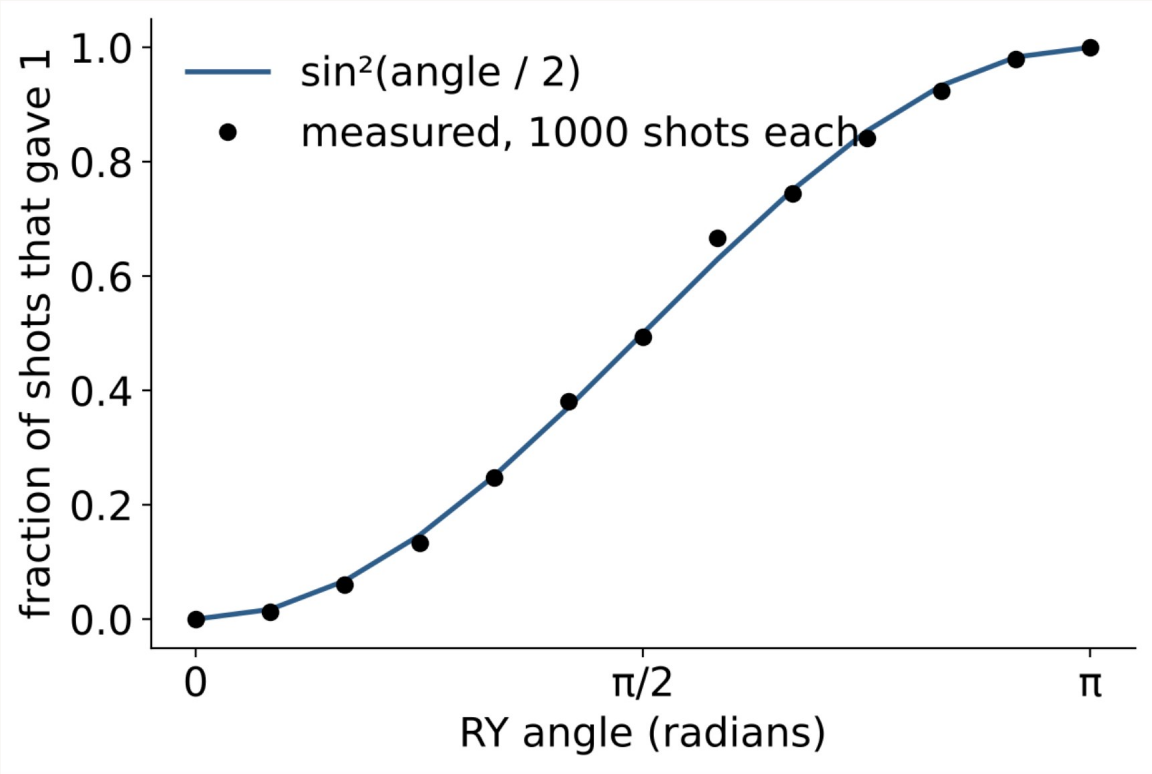
- Amplitudes can be negative. Probabilities cannot.
- The second H sends +0.5 and −0.5 to outcome 1. They cancel. The qubit is 0 for certain.
- Every quantum algorithm works by cancelling wrong answers this way



Exercise 3 in the notebook: three H gates, then four. What is the rule?

RY: choose the odds

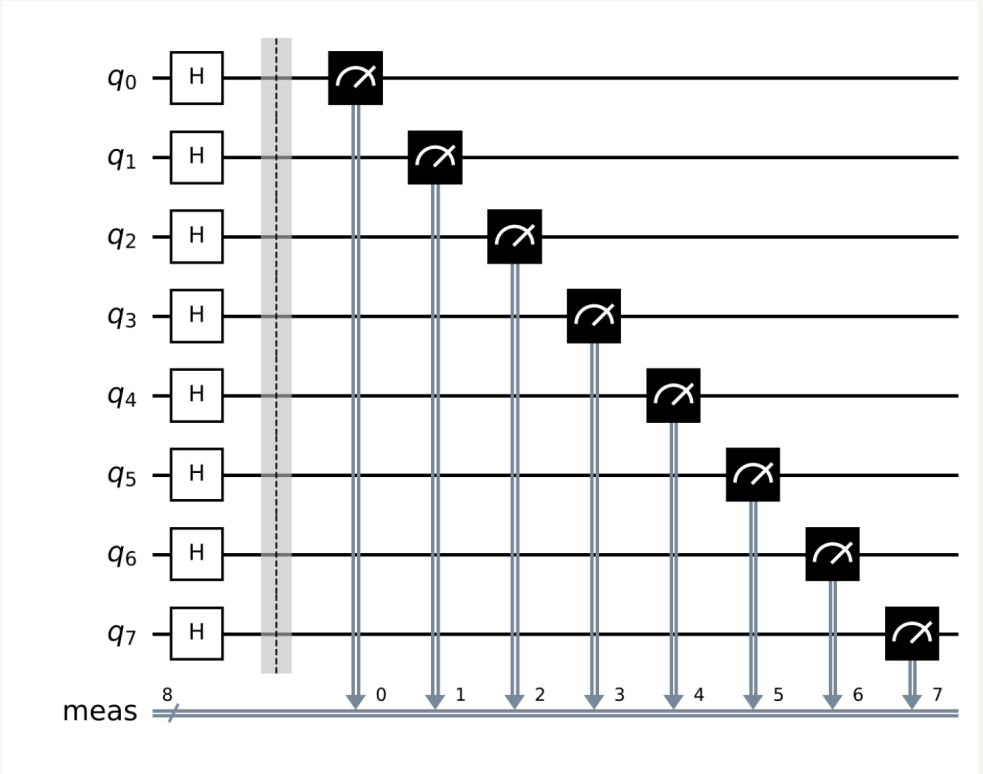
- RY takes an angle. Angle 0 gives always 0. Angle π gives always 1.
- Angle π/2 is fifty-fifty, the same as H
- Probability of 1 is sin²(angle / 2). The dots are 1000 shots at each angle.



Section 4 of the notebook. Exercise 4: find the angle for 25% ones.

More qubits, more bits

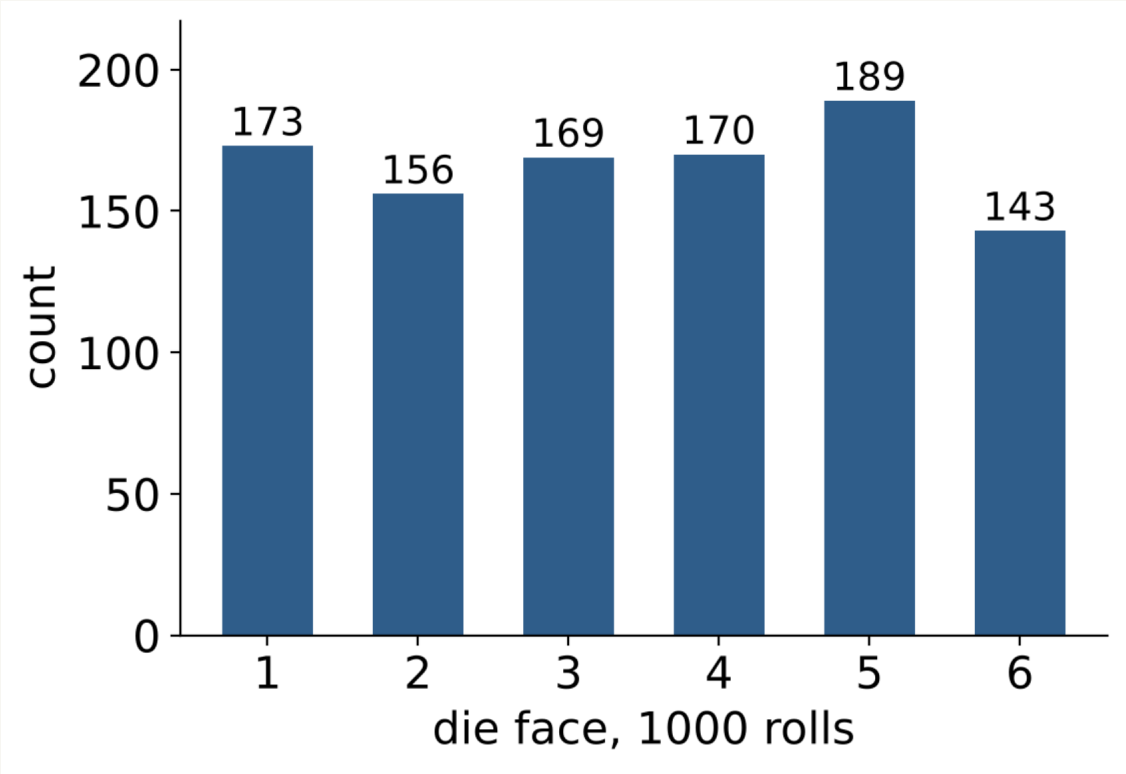
- Eight qubits, H on each, measure all
- One shot gives eight bits: a byte, 0 to 255
- int(bitstring, 2) turns the string of bits into a number



Section 5 of the notebook: memory=True keeps every shot instead of just counting

A quantum die

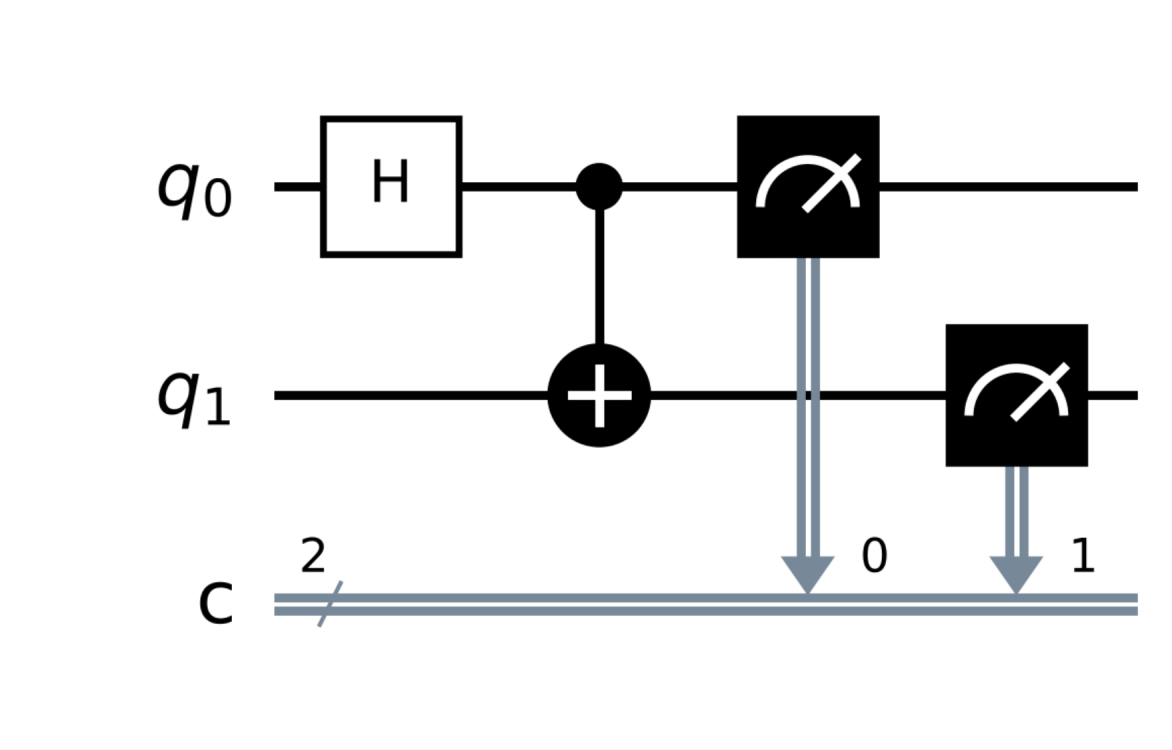
- Three qubits give 0 to 7. A die needs 1 to 6.
- Throw away 6 and 7 and roll again. Every real random library does this.
- Add 1 to what survives. The histogram comes out flat.



Exercise 5 in the notebook: build a d20 from 5 qubits

CX: one qubit controls another

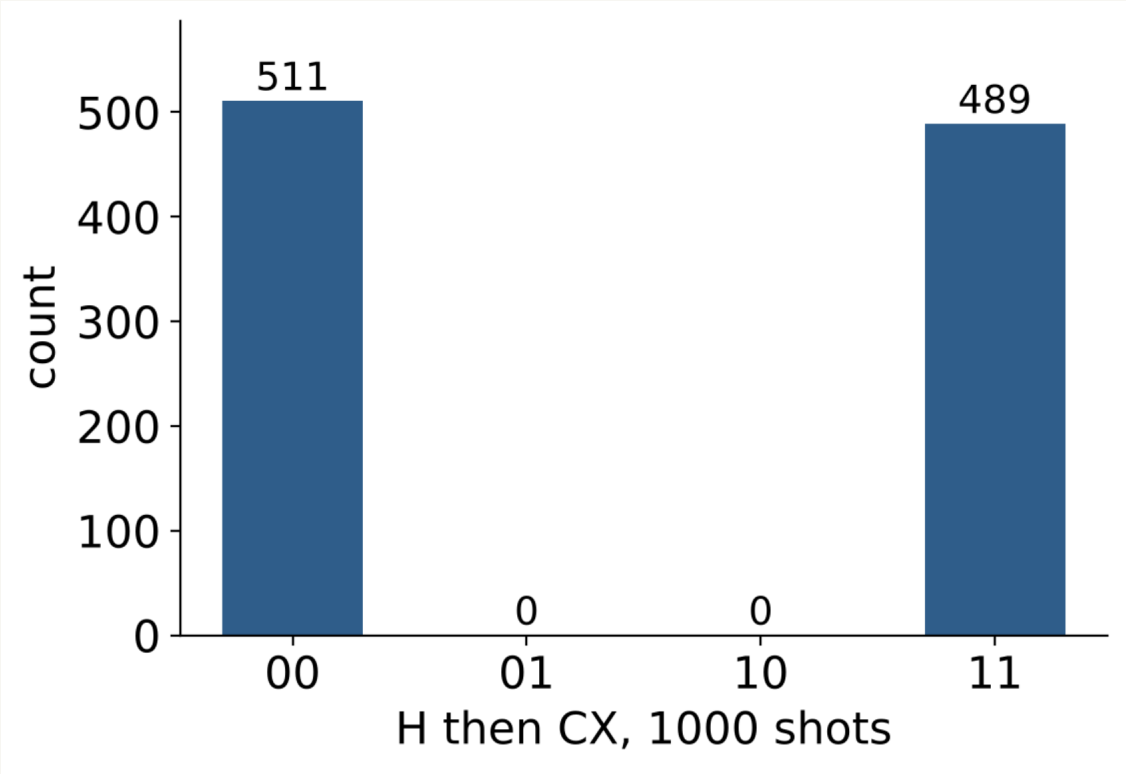
- CX flips the second qubit if the first is 1, and does nothing if it is 0
- On ordinary bits that is just an if statement
- Put H on qubit 0 first, then CX from qubit 0 to qubit 1, then measure both



Section 6 of the notebook: .cx(0, 1)

Entanglement: random alone, locked together

- Each qubit on its own is a perfect random bit
- Together they always agree. Only 00 and 11 ever appear.
- No hidden value was decided in advance and copied. Proving that was the 2022 Nobel Prize in Physics.



Exercise 6 in the notebook: add an X on qubit 1 and predict the two outcomes

What you now know

- A circuit is qubits, gates, and measurements. Measuring gives a bit.
- Amplitudes square to probabilities and can cancel
- Entangled qubits are random alone and locked together

Gate	What it does
H	fifty-fifty bit
RY(angle)	a bit with any odds you choose
X	flips 0 to 1 and 1 to 0
CX(control, target)	flips target if control is 1
measure	reads a qubit into a classical bit

The notebook ends with a table of every Qiskit call used today

Now the notebook

- Open quantum-hello-world on Kaggle and run the cells in order
- Every section ends with an exercise cell. Change the value, predict, run.
- Raise a hand when something does not match the slide

Section	Builds	Minutes
1	classical baseline	5
2	one random bit	10
3	amplitudes and cancellation	10
4	biased bit with RY	10
5	random byte and a die	10
6	entangled pair	10

11:45 hackathon projects. 12:00 lunch.